# Feasibility of Linking Maternal Health Data Without Unique Identifiers in Uganda: A case study of ~22K ANC appointments.

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Mayuge HC IV · Buluba Hospital · 2023–2024

# Introduction

Maternal health data (ANC, delivery, postnatal) are **collected in separate systems**, leading to fragmented and inconsistent records

Lack of **unique identifiers** limits tracking of women across the care continuum

**Probabilistic record linkage** can connect records using variables (e.g., name, age, residence, dates) but has not been well explored in Ugandan settings

**Why this matters:**

- Enables assessment of **continuity of care (ANC → delivery)** and identification of service gaps
- Provides a **low-cost, scalable alternative** in resource-limited settings
- Allows creation of **retrospective and prospective cohorts** from routine data
- Supports **longitudinal maternal health research** without costly primary data collection
- Informs **health system improvement, data integration (EHRs/TREs), and national policy**

# The dataset

| | | | |
|---|---|---|---|
| **~22k** | **2** | **2023-24** | **100+** |
| ANC visit records | facilities | calendar years | variables per record |

## FACILITIES

### Mayuge HC IV

Health Centre IV: primary referral, high-volume ANC clinic

### Buluba Hospital

Faith-based hospital: broader services, longer paper records

**The challenge**

Each row is one visit, but we lack unique patient identifiers. Before any analysis, we need to know who is who.

VARIABLES IN THE DATA

# What every visit record contains

### Identifiers
client number, first/last name, contact, facility, visit date

### Demographics
age, district, subcounty, parish, village, category (new/return)

### Obstetric history
gravida, parity, gestational age, ANC1 timing, ANC visit number

### Anthropometry & vitals
weight, height, MUAC, blood pressure (systolic & diastolic)

### Laboratory
hemoglobin levels, blood group, syphilis, hepatitis, urine

### HIV / PMTCT
EMTCT codes, ART, CD4, infant feeding counselling

### Care actions
tetanus, IPT dose, freellin (ITN), mebendazole, ironacid

### Diagnoses & outcomes
diagnoses, complications, referrals

VARIABLE DEFINITIONS

# Two engineered variables built for linkage

*We derived two new variables from the raw fields to help with potential linkage issues:*

## est_mother_birth_year

### Estimated birth year of the mother

```
year(visit_date) − ageyrs
```

**WHY**

The mother's age could increase between visits but the year she was born would not. Comparing records on mother's birth year means visits by the same mother a year apart should still align. Note: we may still have discrepancies due to birthdate timings.

## numeric_delivery_date

### Expected delivery date (numeric)

```
make_date(year = expected_yrdlvry,
          month = expected_mmdlvry,
          day = delvry_day)
OR
visit_date + weeks(40) - weeks(anc_gestationalage)
```

**WHY**

Built from expected delivery day + month + year when present, or from gestational age otherwise. Converted to a number so it can be compared with a tolerance.

LINKAGE VARIABLES

# Which variables we use to link records — and how

| EXACT MATCH<br>*must be identical* | STRING DISTANCE<br>*tolerates spelling errors* | NUMERIC DISTANCE<br>*small gap = same* |
|---|---|---|
| **client_no**<br>*ANC register number* | **first_name**<br>*Given name*<br>**last_name**<br>*Family name*<br>**client_contact**<br>*Mobile number*<br>**location_combined**<br>*Parish + village* | **est_mother_birth_year**<br>*Visit year − age*<br>**anc_parity**<br>*Previous live births*<br>**anc_parity_2**<br>*Previous pregnancies ≥ 24 weeks stillbirth or miscarriage*<br>**anc_gravida**<br>*Previous pregnancies*<br>**anc_height**<br>*Height (cm)*<br>**delivery_date**<br>*Expected delivery* |

*Each variable contributes one piece of evidence; the model combines them into a single probability of matching per pair.*

THE METHOD

# How the linkage is done

*For every pair of records, the model asks: how likely is it that these two visits belong to the same woman?*

1

### Compare fields

Each variable is compared by the rule we set: exact, string-similarity, or numeric-distance matching.

2

### Score the pair

The fastLink R package combines field-level evidence into a single probability of a match, weighting informative agreements more.

3

### Cluster into patients

Pairs above the probability threshold are linked. Connected records are grouped into patient IDs.

4

### Validate and iterate

We spot-check the results using real cases and adjust our methodology if results are not credible.

PART 01

# Buluba Hospital: Three attempts at linkage

*Each attempt: our methodology, our results, our learnings*

# Initial dataset: Buluba Hospital

**Starting with a more manageable dataset**

We started working with the data from a single hospital to identify data issues on a manageable number of records.

**3,839**

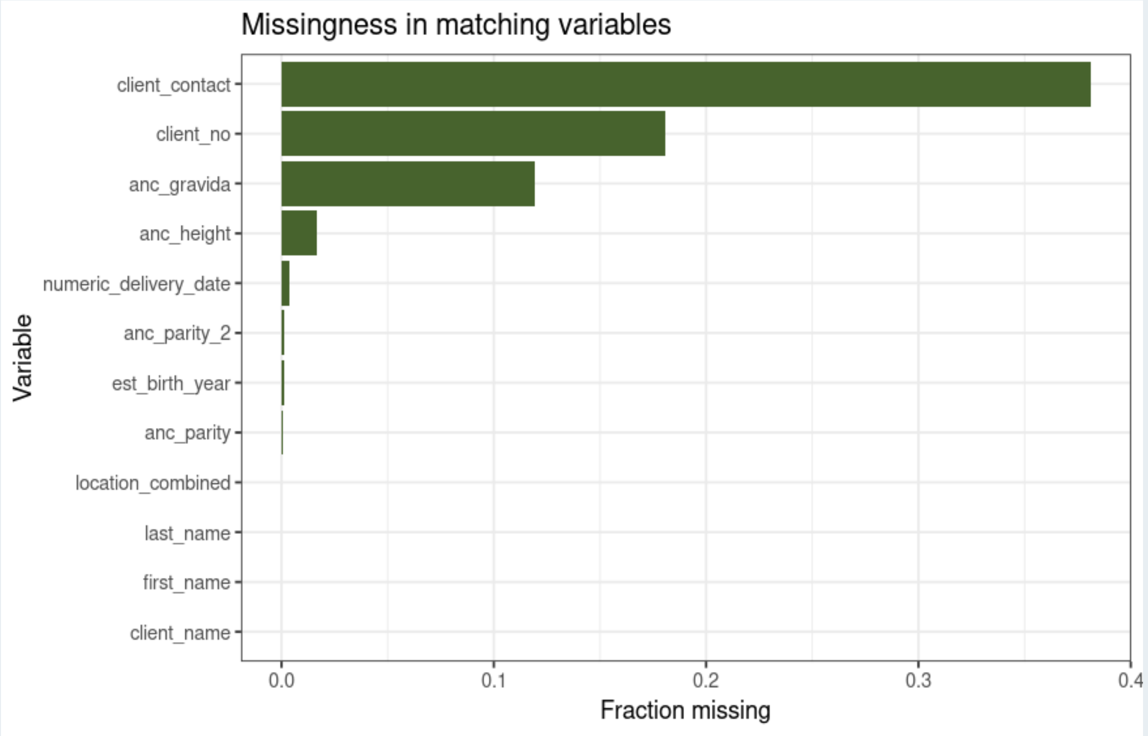
ANC visit records

**38%**

Client contacts missing

**18%**

Client number missing




*Proportion of missingness for matching-relevant variables*

ATTEMPT 1

# First attempt — generous threshold, all variables included

*Start broad: include every variable and accept any pair with a match probability over 85%.*

## WHAT WE DID

- Used all 11 matching variables: client_no, first name, last name, contact, location, est_birth_year, parity, parity_2, gravida, height, delivery date
- Set match threshold to 85% (a relatively permissive cutoff)
- Built patient IDs by grouping all linked record-pairs

## WHAT WE GOT

UNIQUE PATIENTS

**497**

LARGEST CLUSTER

**2,724**

**Too lenient — clearly broken**

*A single ‘patient’ with 2,724 visits is impossible. Far too many distinct women have been lumped together.*

ATTEMPT 2

# Second attempt — same variables, stricter threshold

*We increased the threshold for a match to 97% probability in an attempt to break up the giant cluster.*

## WHAT WE DID

- Kept all 11 matching variables — no changes to the variable set
- Raised the match threshold from 85% to 97%
- Re-ran fastLink and re-clustered the resulting pairs

## WHAT WE GOT

| UNIQUE PATIENTS | LARGEST CLUSTER |
|---|---|
| 1,191 | 534 |

**Better — but chains persist**

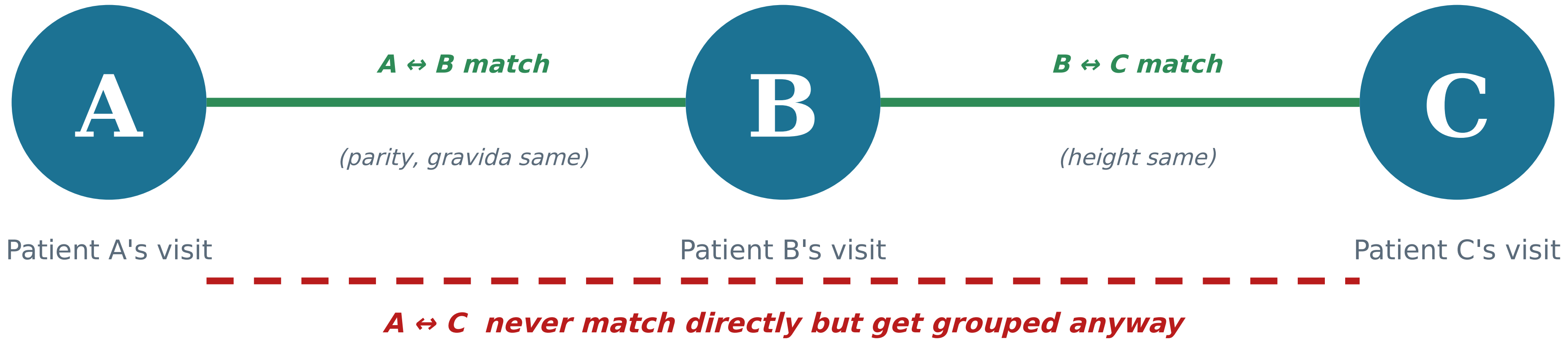
*More patients identified, biggest cluster has shrunk by 80%. But 534 visits in one 'patient' is still impossible: the threshold alone is not enough.*

DIAGNOSIS

# Why a giant cluster forms: ‘friend of a friend’

*Matching process:*

- *Pairwise matches get grouped together*
- *Clusters of records that are linked share a patient ID.*

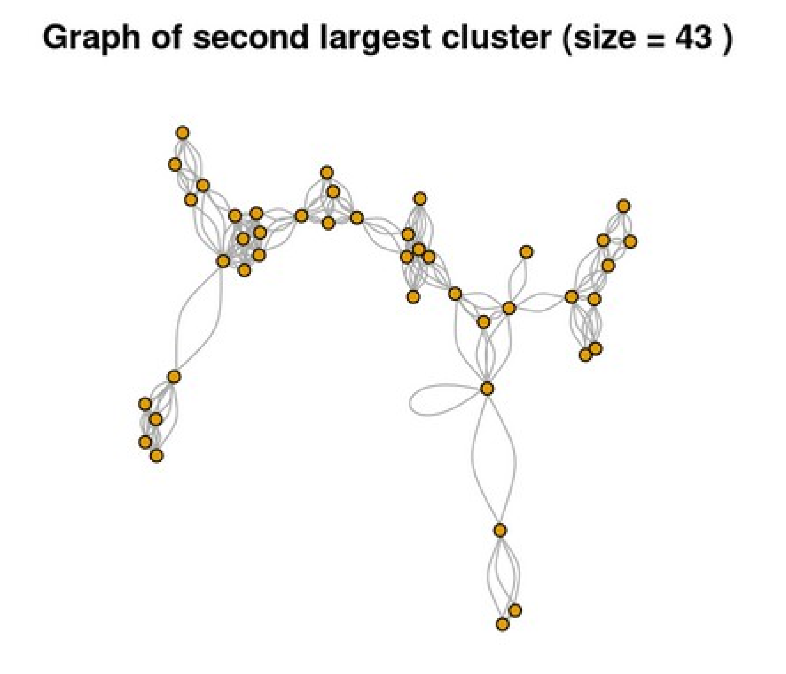



**‘Transitive closure’**: Common values (e.g. parity = 2 and gravida = 3), or unreliable values with limited ranges (e.g. height) chain unrelated women together through intermediates. Tightening the threshold alone will not solve this.

DIAGNOSIS

# What the chains look like inside a cluster

*Each dot is one ANC visit; each line is a match. A real cluster of repeat visits should be tightly interconnected.*




*Second-largest cluster (43 records, threshold 0.97)*

## THE FIX

Long chains form on weak evidence: matches on variables many women share. The fix is not a higher threshold but a smaller, more distinctive variable set.

## DROP

`anc_parity, anc_parity_2, anc_gravida, anc_height`

ATTEMPT 3

# Third attempt — stricter threshold and fewer variables

*Keep the 97% threshold; drop four variables with common values or messy data.*

## WHAT WE DID

- Kept threshold at 0.97
- Dropped anc_parity, anc_parity_2, anc_gravida, anc_height
- Retained client_no, names, contact, location, est_birth_year, numeric_delivery_date — the seven most individuating fields

## WHAT WE GOT

| UNIQUE PATIENTS | LARGEST CLUSTER |
|---|---|
| 1,759 | 10 |

**Plausible — adopted**

*Patient count increased by 550+ and the largest cluster is now 10 visits — clinically realistic for a high-volume ANC client over two years.*

VALIDATION

# Validation — client 189 splits cleanly into 4 patients

*The same client number (189) was used by multiple women. The final model assigns four separate patient IDs , matching the manual reading of the records.*

**PATIENT #387**

***Patient A***

**1 visit**

*Distinct name and earliest delivery date by almost 1 year*

**PATIENT #341**

***Patient B***

**5 visits**

*Slight spelling drift in names and contact number correctly merged*

**PATIENT #201**

***Patient C***

**1 visit**

*Distinct name*

**PATIENT #41**

***Patient D***

**3 visits**

*Younger patient with matching name and consistent contact number (when present).*

✓ ***Four distinct women, correctly separated***

# Validation — are we now too strict?

*We do not want to overcorrect and separate appointments that should be identified as belonging to the same patient.*

| Assigned ID | Visit date | Client Number | Name | Contact | Estimated birth year | Height & parity | Delivery date | Parish & village |
|---|---|---|---|---|---|---|---|---|
| Patient #1379 | 2024-11-21 | 139 | Off by 1 letter | [missing] | Matching | Matching | 25 days apart | Lukone, Lukone |
| Patient #1711 | 2024-12-19 | 329 | | | | | | Katonte, Katonte |

***✗ Likely incorrectly separating appointments belonging to the same person***

SUMMARY

# The journey at a glance

**ATTEMPT 1**

*All variables · threshold 0.85*

UNIQUE PATIENTS

**497**

BIGGEST CLUSTER

**2,724**

**Too lenient**

**ATTEMPT 2**

*All variables · threshold 0.97*

UNIQUE PATIENTS

**1,191**

BIGGEST CLUSTER

**534**

**Improved — chains remain**

**ATTEMPT 3**

*Distinctive variables · threshold 0.97*

UNIQUE PATIENTS

**1,759**

BIGGEST CLUSTER

**10**

**Plausible — adopted**

PART 02

# Two hospitals: total linkage attempt

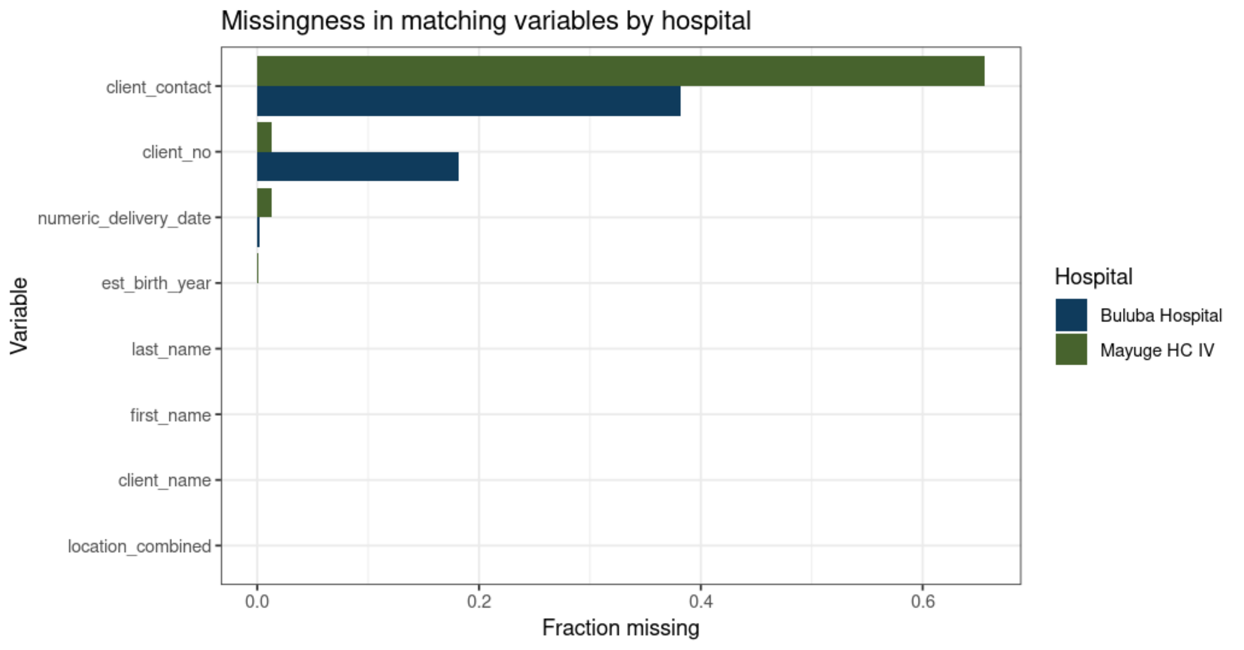
*Linkage on Mayuge HC IV and Bulubu Hospital*

INTRODUCTION TO THE DATA

# Total dataset: Mayuge HC IV & Buluba Hospital

## Complete dataset

We now have 22,037 visit records: 3,839 from Buluba and 18,168 from Mayuge, a 5x increase in number of visits from our first linkage attempts.




*Mayuge HC IV has high missingness in client contact (66%), while Buluba has higher missingness in client number (18%).*


*Significant discrepancy in the entered expected delivery date and the expected delivery date based on gestational age*

ATTEMPT 1

# First attempt — fewer variables

*Including the variables that gave us the best success on the Buluba data, using the 85% match threshold.*

## WHAT WE DID

- Used 7 matching variables: client_no, names, contact, location, est_birth_year, numeric_delivery_date
- Set match threshold to 0.85
- Built patient IDs by grouping all linked record-pairs (graph components)

## WHAT WE GOT

| UNIQUE PATIENTS | LARGEST CLUSTER |
|---|---|
| 9,497 | 11 |

**Plausible**

*Plausible value for a maximum number of appointments for a single patient*

# Validation

*With the new, larger dataset, client number 189 now splits into 8 patients, matching manual review:*

## BULUBA HOSPITAL

### 4 patients

- Patient A: 1 visit
- Patient B: 5 visits
- Patient C: 1 visit
- Patient D: 3 visits

## MAYUGE HC IV

### 4 patients

- Patient E: 1 visit
- Patient F: 2 visits
- Patient G: 1 visit
- Patient H: 4 visits

***✓ Eight distinct women, correctly separated***

*But we have the same problem with the individual with 2 appointments still being misidentified as 2 different patients:*

<table>
<tr><th>Assigned ID</th><th>Client Number</th><th>Name</th><th>Estimated birth year</th><th>Height & parity</th><th>Delivery date</th><th>Parish & village</th></tr>
<tr><td>Patient #8733</td><td>139</td><td rowspan="2">Off by 1</td><td rowspan="2">Match</td><td rowspan="2">Match</td><td rowspan="2">25 days apart</td><td>Lukone, Lukone</td></tr>
<tr><td>Patient #7064</td><td>329</td><td>Katonte, Katonte</td></tr>
</table>

***✗ Likely incorrectly separating appointments belonging to the same person***

PART 03

# Baseline descriptive statistics

*Who are the 9,497 women we now have in the cohort?*

DEMOGRAPHICS

# Maternal age

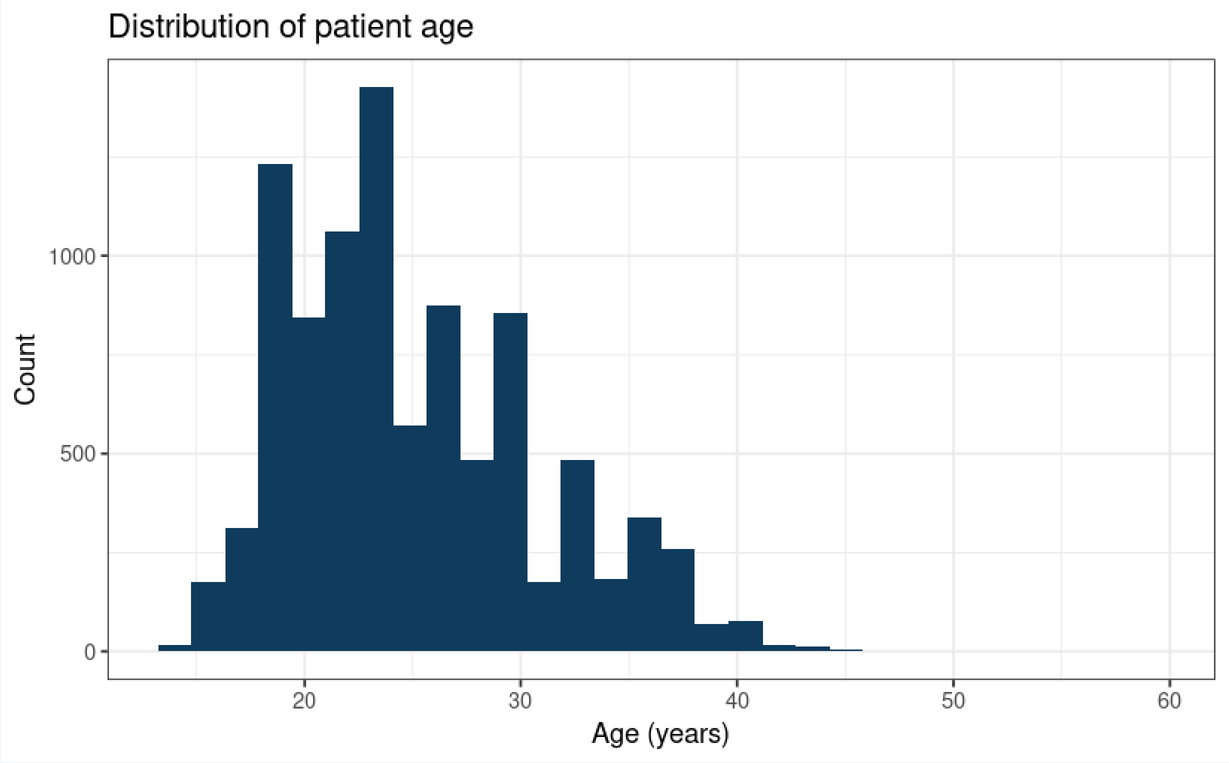
| MEDIAN AGE | IQR |
|---|---|
| **24** | **20–29** |
| *years* | *years* |

| MEAN (SD) | RANGE |
|---|---|
| **25.0** | **14–59** |
| *(5.9)* | *years* |

COMPLETENESS OF AGE VARIABLE

**>99.9%**

*complete*




*Distribution of age at time of appointment*

GEOGRAPHY

# Where the women come from

**84%**
live in Mayuge district

**14%**
lack a district at all

**49**
districts represented

## COMMON PARISHES OF MAYUGE

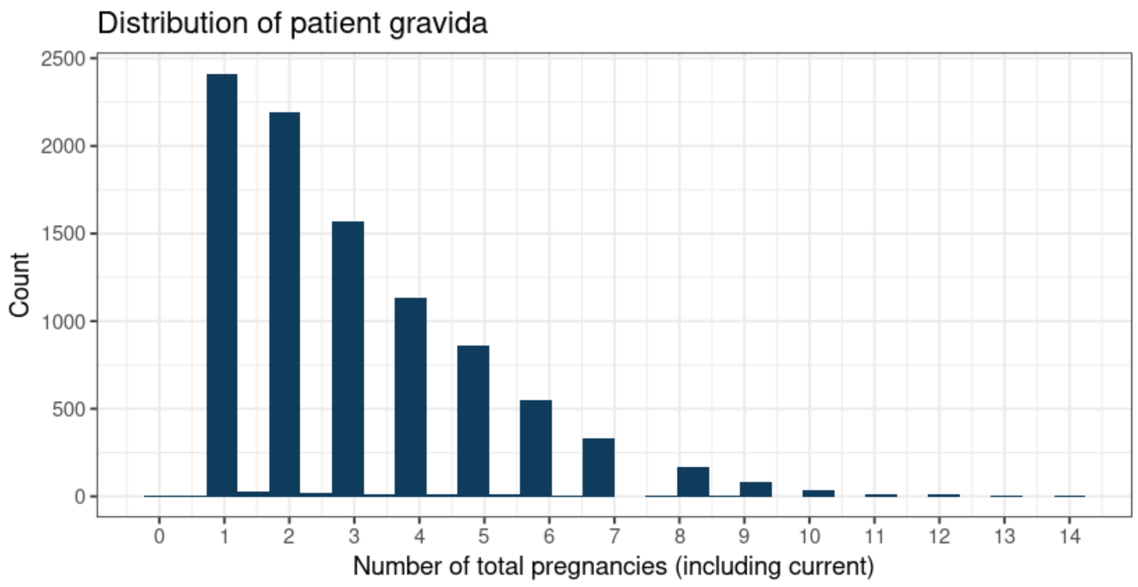
Top 5:
- Kasugu: 826 patients
- Lugolole: 588 patients
- Buwolya: 518 patients
- Ikulwe: 416 patients
- Maina: 401 patients

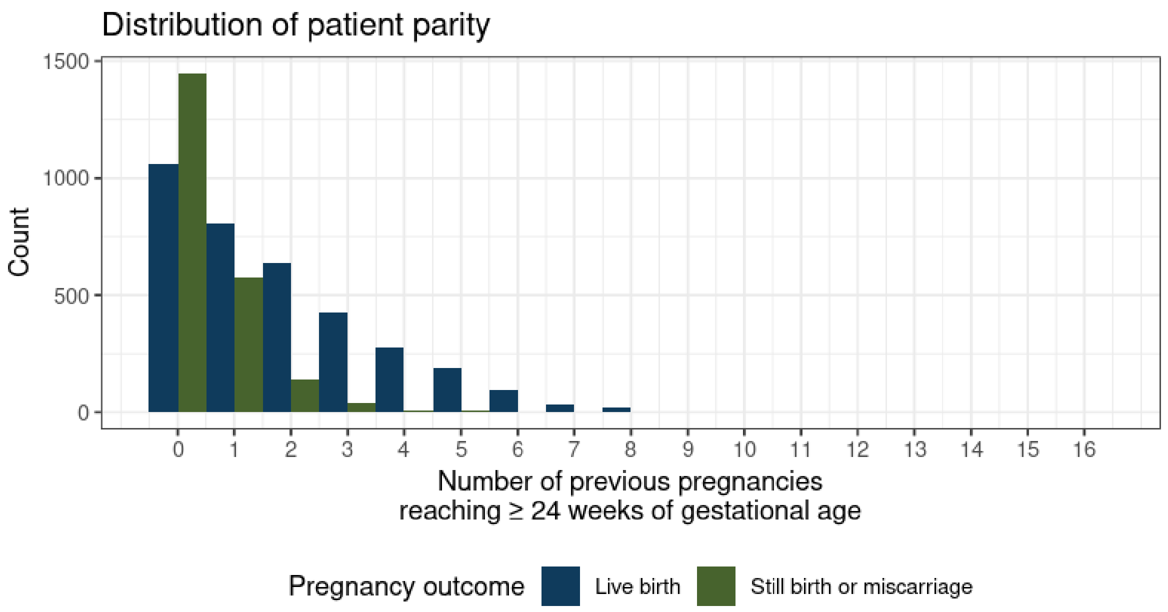
## OTHER DISTRICTS REPRESENTED

Top 5:
- Jinja: 41 patients
- Iganga: 30 patients
- Bugweri: 18 patients
- Luuka: 18 patients
- Kampala: 17 patients

# Obstetric profile

| MEDIAN GRAVIDA | PRIMIGRAVIDA | MEDIAN PARITY |
|---|---|---|
| **3** | **25%** | **1+0** |
| *(IQR 2–5)* | *first pregnancy* | *(63% missing data)* |

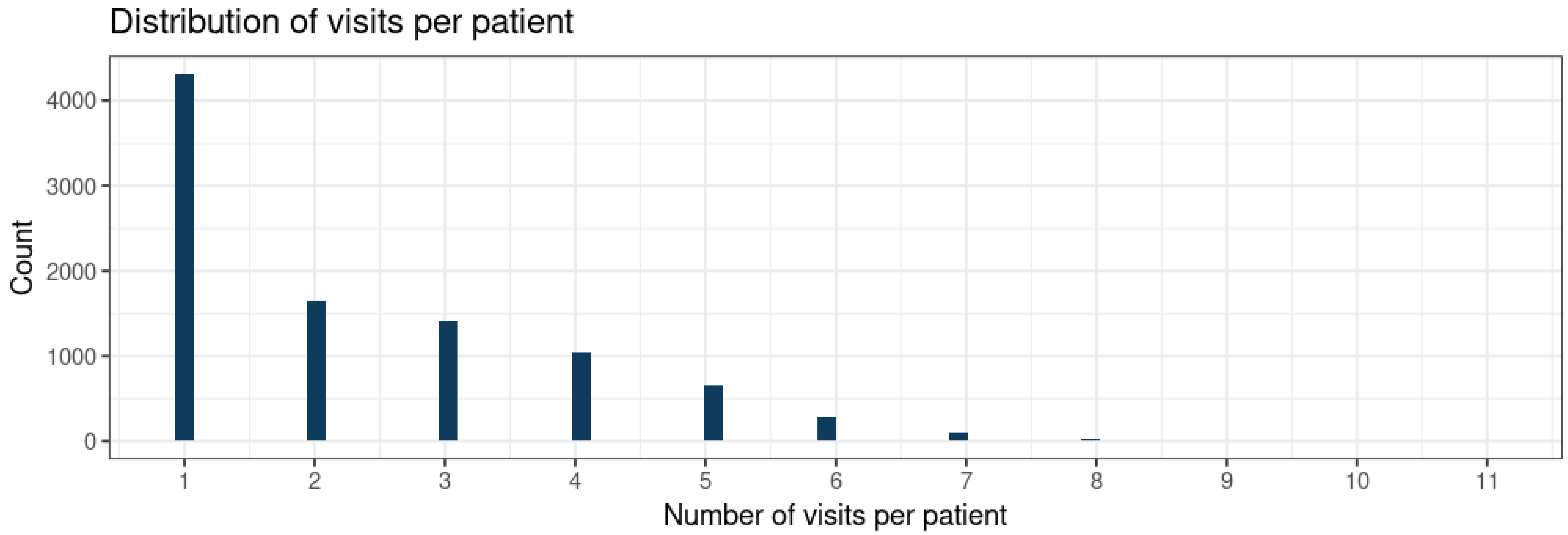


ANC ENGAGEMENT

# How many times each woman attended

| MEDIAN VISITS | SINGLE-VISIT | ANC4+ | MAX OBSERVED |
|---|---|---|---|
| **2** | **45%** | **22%** | **11** |
| *(IQR 1–3)* | *attended once* | *completed 4+ visits* | *visits in window* |



FROM VISITS TO PATIENTS — AND ONWARDS

# 9,497 women now identifiable across all 22,037 visits.

*With the linkage in hand, we can now link forward to delivery and postnatal datasets, investigate maternal and child outcomes, and build the first AI-ready perinatal cohort from this catchment.*

Questions and discussion